Opacity as Commitment: Fragile Equilibria under Delegated Trading*

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Abstract

We study a model à la Kyle (1985) in which financial traders care not only about profits but also their reputation, i.e., a recruiter's perception of their skills. The model presents two self-fulfilling equilibria. One resembles Kyle's original equilibrium. In the other, a "reputation-driven" equilibrium, traders trade more aggressively to appear skilled, leading to poorer performance and higher price volatility. We show that (1) labor mobility in finance generates traders' reputation concerns, triggering market fragility and increasing wage inequality, (2) traders with different skill levels adopt different trading styles, and (3) skills in active management may not translate into performance.

JEL classification: G11, G12, G14, G23

Key words: informed trading, market reputation, trader skill, financial fragility, poaching, asset prices

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1 Introduction

Financial firms compete for skilled employees. As the boundaries between investment banking, commercial banking, and asset management blur, this competition intensifies. Following the 2007-2009 crisis, pressures to reduce compensation and the resulting challenges in retaining talent have brought poaching practices into focus.[1] Poaching, or more generally labor mobility, raises concerns among financial specialists about their reputation within the industry. How do these concerns influence their trading strategies and skill acquisition? How do they interact with asset prices and market liquidity?

To address these questions, we study a model inspired by Kyle (1985), where traders, privately informed about their skill, trade risky assets anonymously with market makers. We define skill as the trader-specific ability to uncover more information about asset payoffs than the market. The more skilled a trader is, the larger the potential mispricing she can identify, and thus, the greater the trading profits she can potentially earn. Each trader selects an asset based on her skill and strategically submits a market order to the market maker for that asset, who sets the price after observing the aggregate order flow. A recruiter attempts to infer each trader's skill from the observed price and payoff of the asset chosen by the trader. The trader values not only trading profits but also her reputation, i.e., the recruiter's estimate of her skill. In a simple one-period model presented in Section 2, each trader derives an exogenous benefit from her reputation, which we model as an increasing quadratic function of reputation. In Section 4, we extend the model to a two-period setting, providing a microfoundation for the reputation benefit. In this extension, each trader randomly encounters a recruiter who compensates her for her skill based on his estimate. In equilibrium, the quadratic form assumed

[1] According to *The Wall Street Journal*, poaching is a common practice in the sophisticated capital markets industry, occurring in both bull and bear markets ("Greed is Good," February 2009), and employees at all levels are being poached ("Employee Poaching Returns to Wall Street," May 2010). *The New York Times* ("A Mad Scramble for Young Bankers," July 2014) and *Bloomberg* ("Saturdays Off Lose to Cash as Buyout Firms Poach Junior Bankers," March 2014) highlight the buy-side's (private equity, hedge funds) recruitment of young talent from investment banking amid a changing financial industry.

in Section 2 arises endogenously.

The model yields two stable, self-fulfilling equilibria, one of which is similar to that of Kyle (1985), where reputation is irrelevant. In the other, the "reputation-driven" equilibrium, traders trade more aggressively to appear skilled. This strategy increases trading volume, price volatility, and market liquidity but undermines trading profits by revealing too much private information to the market maker. Intuitively, each trader can boost her reputation by placing a large order, either buy or sell, because such an action makes her appear like a skilled trader who has identified an asset with significant mispricing. In equilibrium, the recruiter accurately anticipates the trader's strategy, so the recruiter's learning is not manipulated; nevertheless, the trader may trade aggressively depending on the market's belief. To illustrate the mechanism, suppose the recruiter believes that the trader trades more aggressively than the profit-maximizing level. If she trades less aggressively to increase profits, the recruiter (incorrectly) infers that she is less skilled. To avoid the reduction in her reputation benefit, she conforms to the recruiter's belief and indeed trades very aggressively.

The self-fulfilling property of our equilibria implies that a mere shift in investor beliefs, unaccompanied by changes in fundamentals, can trigger a transition from the standard Kyle-like equilibrium to the reputation-driven one, where price volatility is high and trading performance is relatively poor. We interpret this as *fragility* in financial markets.[2] Key to our multiple equilibria and fragility results are the trader's conflicting motives for revealing skill: on one hand, she wishes to hide her skill from the market maker to earn trading profits, but on the other hand, she wants to show it off to the recruiter to obtain the reputation benefit. If the trader's concern for reputation is very weak (very strong), the first (second) motive dominates, and the Kyle-like (reputation-driven) equilibrium holds uniquely. However, if the concern is moderately strong, either equilibrium may arise depending on the market's self-fulfilling beliefs. Therefore, the model predicts that fragility may occur

[2]As in Greenwood and Thesmar (2011), we define a market as fragile if it is susceptible to non-fundamental demand shifts, such as those caused by changes in investor beliefs.

when poaching is (moderately) active and there is high demand for financial skill, as these situations increase the likelihood of future recruitment and lead to heightened reputation concerns.

Our results shed light on how traders' underlying skills relate to their trading styles and contribute to financial fragility. The model predicts that high-skilled traders adopt low-intensity, profit-driven strategies, while those with lower skills engage in aggressive, risk-seeking behavior to gain reputation. Traders with intermediate skill levels, facing multiple equilibria, may switch between profit-seeking and reputation-driven strategies based on market beliefs, thereby contributing to market fragility. This behavior aligns with real-world observations during financial crises, such as momentum crashes and sudden reversals, which occur even without fundamental shocks.

The model has implications for compensation in the finance industry. In the reputation-driven equilibrium, traders' wages exhibit greater inequality among high- and low-skilled traders compared to those in the Kyle-like equilibrium, as they trade aggressively and reveal their skill more than they do in the Kyle-like equilibrium. This allows the recruiter to better assess each trader's skill and, therefore, to offer wages that more accurately reflect their abilities. This result is consistent with the findings by Philippon and Reshef (2012) and Célérier and Vallée (2019), who document increasing returns to talent and their significant contribution to income inequality in the finance industry.

The aggressive trade in the reputation-driven equilibrium also causes traders to reveal excessive information to the market maker, resulting in poorer trading performance than in the Kyle-like equilibrium. Consequently, high-skilled traders in the reputation-driven equilibrium may underperform low-skilled traders following the Kyle-like equilibrium. This result offers a novel explanation for the puzzling relationship (or the lack thereof) between trader skill and persistent performance, as documented and analyzed in the literature (e.g., Busse, Goyal, and Wahal, 2010; Berk and Green, 2004).

In Section **??**, we study traders' endogenous skill acquisition. We show that, due to the multiplicity of equilibria, each trader may intentionally acquire a “middling” level of skill, which is too high or too low compared to the level she would acquire if she knew which type of equilibrium would prevail. Ex-post, in the Kyle-like equilibrium, where profits are prioritized over reputation, the acquired skill level is insufficient. Conversely, in the reputation-driven equilibrium, where reputation outweighs profit motives, the skill level is excessive. Ex-ante, without knowing which equilibrium will emerge, the trader optimally chooses a middling skill level to prepare for both scenarios. This finding aligns with real-world observations that some finance professionals tend to pursue broader business qualifications rather than specializing in areas like advanced computer science or pursuing higher academic degrees such as a PhD.

Our paper is related to the literature on fragility in financial markets. Cespa and Vives (2015) is most closely related, as they also find multiple equilibria in an asymmetric information trading model. The key ingredients of their model are short-term investor horizons and persistent liquidity trading, which generate strategic complementarities. In Froot, Scharfstein, and Stein (1992), strategic complementarities among short-term traders are also important for the emergence of multiple equilibria. Unlike these studies, our model does not feature short horizons; instead, we obtain fragility from traders' forward-looking concerns about future recruitment opportunities. In Cespa and Foucault (2014), learning from other assets amplifies shocks and causes liquidity crashes. Although the mechanism is different, our model can also result in liquidity crashes when the economy switches between equilibria due to non-fundamental shocks.

Our paper also connects to the literature on trading frenzies, such as Veldkamp (2006) and Goldstein, Ozdenoren, and Yuan (2013), in which strategic complementarities play a key role. In Veldkamp (2006) frenzies originate from complementarities in information markets, while in Goldstein, Ozdenoren, and Yuan (2013) strategic complementaries, triggered by a feedback effect from financial markets to a firm's real investment, lead to coordinated aggressive trading. By contrast, in our reputation-driven equilibrium, each trader trades aggressively out of fear of being perceived as less skilled by the recruiter who believes that she would trade aggressively.

Lastly, our work contributes to the broader understanding of the role of asymmetric information about skills in financial markets. DiMaggio (2015), Malliaris and Yan (2021), and Bijlsma, Boone, and Zwart (2018) derive implications for the risk-taking behavior of agents when ability is private information but is reflected in performance. We also link trader skill to private information and asset markets; however, unlike these papers, our focus is on deriving pricing implications. Guerrieri and Kondor (2012) show that career concerns amplify price volatility, similar to our findings. In Dasgupta (2006), career concerns lead to churning (trading without private information), and in Dasgupta and Prat (2008), they lead to herding (ignoring private information and trading like everyone else). These three papers are closely related to ours in that they study career concerns and endogenous asset prices, but they do not analyze fragility, which is the main focus of our paper.

The rest of the paper proceeds as follows. Section 2 presents a baseline model with exogenous reputation concerns, and Section 3 studies the equilibrium. Section 4 endogenizes reputation concerns by introducing a two-period model and presents the implications of our results. Section **??** analyzes traders' skill acquisition. The Online Appendix contains all proofs for the theoretical results.

2 Model

This section presents a one-period trading model à la Kyle (1985) with delegated investments. The economy consists of a risk-neutral *fund manager*, a competitive risk-neutral *market maker*, *noise traders*, and $n \geq 2$ risk-neutral *investors*. The manager leverages her skill (defined later) to discover a single risky asset with potential mispricing, raises funds from investors, and trades the asset with the market maker who set the price p. The remaining funds are allocated to risk-free savings with a zero interest rate.[3]

[3]We abstract away from information acquisition by individual investors. Economies of scale would support this assumption as a natural equilibrium outcome, in which investors do

Skill. The manager can uncover more information about the asset payoff than the market maker, and we interpret it as her skill as follows. At the beginning of the period, the manager draws a number s from a normal distribution, $N(0, \omega_s)$, where s is her private information. Leveraging s, she identifies an asset whose payoff is $\delta = \bar{\delta} + s$ with a publicly know mean payoff $\bar{\delta}$. Since the market maker does not observe s, the prior mean of δ from their perspective is $\mathrm{E}[\delta] = \bar{\delta} + \mathrm{E}[s] = \bar{\delta}$ and is biased by s from the manager's perspective. As the market maker would set a price $p = \mathrm{E}[\delta] = \bar{\delta}$ in the absence of traders, s can be viewed as the asset's "potential mispricing."[4] Indeed, as shown later, the manager profits by buying the asset if $s > 0$ (underpriced) and selling it if $s < 0$ (overpriced); the larger the $|s|$, the larger the profit. Moreover, the variance $\mathrm{Var}[s] = \omega_s$ controls the shape of the underlying distribution of s: a large ω_s represents a wider scope, making the manager more likely to identify an asset with a large mispricing, $|s|$. In light of this, ω_s is referred to as the manager's asset-selection *skill* with $|s|$ being its realized value.[5] In practical terms, ω_s can reflect factors such as the manager's investment in higher education in finance and economics, which broadens the range of information available to her, while $|s|$ captures the realized ability to leverage such investments into superior decision-making in financial markets. In what follows, when there is no risk of confusion, we may refer to $|s|$ simply as "skill" without explicitly calling it "realized skill."

Investment. We model the fundraising and trading stages as a simultaneous game with rational expectations. On the one hand, the manager decides on the investment into the risky asset, anticipating the investors' and the market maker's reactions. This action is represented by a market order for x units of

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not acquire the necessary skills and information, but rather remain uninformed and invest through a manager (e.g., Gârleanu and Pedersen, 2018). They do not directly participate in the market due to the lack of informational advantages over the market maker.

[4] s is not the *actual* mispricing because the market maker will take into consideration the presence of the informed manager; thus, p will deviate from $\bar{\delta}$ in equilibrium.

[5] This formalization of skill aligns with findings in the empirical studies that attribute fund managers' performance to stock-picking skills (e.g., Wermers, 2000; Kosowski, Timmermann, Wermers, and White, 2006).

the asset sent to the market maker. In the financial market, noise traders also submit random order $u \sim N(0, \omega_u)$ exogenously. The market maker observes the aggregate order flow, $x+u$, and sets the semi-strong efficient price to earn zero expected profit:[6]

$$p = \mathrm{E}[\delta | x+u]. \tag{2.1}$$

In raising funds from investors, the manager disseminates a noisy signal about her position to investors, $x+e$, where e represents a normally distributed noise, $e \sim N(0, \omega_e)$. This signal is delivered through private communication between the manager and her client investors and could capture the fund manager's practice of communicating a noisy signal about a high-conviction, concentrated risky position to investors to attract capital, consistent with the empirical evidence and theoretical rationale in Anton, Cohen, and Polk (2021) and Van Nieuwerburgh and Veldkamp (2010). It is informative about the manager's skill and the return of the fund, and investors use it in deciding on fund allocations. The noise variance, ω_e, is interpreted as the *opacity* of the fund. We first consider the baseline model with a perfectly transparent fund ($\omega_s = 0$), while Section XXX deals with general opacity ($\omega_s > 0$). Section XXX endogenizes ω_s through the manager's choice, e.g., she may obfuscates information by making trading strategies complex and incomprehensible to uninformed investors. Upon observing the fund information, $x+e$, investor i allocates m_i dollars to the fund, so that the fund's asset under management (AUM) becomes $M = \sum_{i=1}^{n} m_i$. The fund's fee structure is proportional to the total profits: the manager takes ϕ fraction of the total fund profit, while the remaining $1-\phi$ fraction is distributed to individual investors according to their contributions to the fund, m_i/M.[7]

[6] Assume that, on the equilibrium path, one competitive market maker trades the asset. As in Kyle (1985), competitively many market makers exist off the equilibrium path, ensuring a break-even price in the equilibrium.

[7] Introducing a fixed fee does not change our qualitative results, as long as the performance-based fee also exists, $0 < \phi < 1$.

Maximization problem. With the trading profit from the risky investment being $\pi \equiv (\delta - p)x$, the total fund proceeding is expressed as

$$y = \pi + M. \tag{2.2}$$

The fund manager maximizes her fee income by controlling her investment strategy:

$$\max_x \mathrm{E}\,[\phi y|\, s]. \tag{2.3}$$

Conversely, investor i maximizes her return from investment by controlling her fund contribution:

$$\max_{m_i} \mathrm{E}\left[\frac{m_i}{M}(1-\phi)y - m_i|x+e\right]. \tag{2.4}$$

Definition of equilibrium. Let $p(x+u)$ represent the market maker's pricing strategy, $x(s)$ denote the manager's trading strategy, and $\{m_i(x+e)\}_{i=1}^n$ denote investors' fund allocations. An equilibrium consists of $p(\cdot)$, $x(\cdot)$, and $\{m_i(\cdot)\}_{i=1}^n$ such that $p(\cdot)$ satisfies (2.1) given $x(\cdot)$, $x(\cdot)$ solves (2.3) given $p(\cdot)$ and $\{m_i(\cdot)\}_{i=1}^n$, and each $m_i(\cdot)$ solves (2.4) given $p(\cdot)$ and $x(\cdot)$.

3 Equilibrium

In this section, we search for a linear equilibrium under transparency ($\omega_s = 0$), in which each trader's market order x is linear in her skill, s. Proofs for the analytical results are provided in the Online Appendix.

3.1 Trading Strategy

Conjecture and later verify that the manager's equilibrium trading strategy is

$$x(s) = \beta s, \tag{3.1}$$

where $\beta > 0$ is an endogenous constant determined in the equilibrium. Conjecture (3.1) states that the manager buys (sells) the asset if her s is positive (negative), where the order size is proportional to her skill $|s|$; the more skilled the manager, the more aggressively she trades.[8] The scale factor β is referred to as the *trading intensity*.

3.2 Execution Price

The market maker decides on the price of the asset by extracting information about the asset's payoff from the aggregate order flow, $x+u$. Anticipating the trading strategy (3.1), the price in equation (2.1) can be rewritten as

$$p = \bar{\delta} + \lambda(x+u), \tag{3.2}$$

where

$$\lambda \equiv \frac{\beta\omega_s}{\beta^2\omega_s + \omega_u} \tag{3.3}$$

is "Kyle's lambda," a measure of the trader's price impact.[9] Following the literature, we define a market to be *liquid* if λ is small.

3.3 Delegation under Full Transparencyd

By applying $y = \pi + M$, investor i's optimization problem in (2.4) is rewritten as

$$\max_{m_i} \frac{m_i}{M}(1-\phi)\mathrm{E}\left[\pi|x+e\right] - \phi m_i, \tag{3.4}$$

leading to the following optimal delegation:

$$m_i = \left(\frac{1-\phi}{\phi}\mathrm{E}[\pi|x+e]\sum_{j\neq i} m_j\right)^{\frac{1}{2}} - \sum_{j\neq i} m_j. \tag{3.5}$$

[8] This conjecture follows from the original Kyle model where an informed trader trades based on the difference between her belief and that of the market maker, i.e., $\mathrm{E}[\delta|s]-\mathrm{E}[\delta] = s$.

[9] From (2.1), we have $p = \mathrm{E}[\delta|x+u] = \mathrm{E}[\mathrm{E}[\delta|x+u,s]|x+u] = \mathrm{E}[\bar{\delta}+s|x+u]$. Since the market maker believes that the manager follows (3.1), s and $x+u$ are both normally distributed from the market maker's perspective. Thus, $p = \bar{\delta} + \mathrm{E}[s] + \frac{\mathrm{Cov}[s,x+u]}{\mathrm{Var}[x+u]}(x+u-\mathrm{E}[x+u]) = \bar{\delta} + \frac{\mathrm{Cov}[s,\beta s+u]}{\mathrm{Var}[\beta s+u]}(x+u-\mathrm{E}[\beta s+u]) = \bar{\delta} + \frac{\beta\omega_s}{\beta^2\omega_s+\omega_u}(x+u)$.

The cost of delegating investment is linear in the dollar value of the stake, as represented by the second term of (3.4), while the marginal benefit of an additional stake is diminishing, as is captured by the first term. Hence, given the total fund allocation by other investors, $\sum_{j \neq i} m_j$, investor i's optimal delegation is determined by the FOC in equation (3.5).

We focus on the symmetric equilibrium where all investors provide the same amount of capital to the fund. By imposing $m_i = m_j$ for all $i, j \in \{1, 2, \cdots, n\}$, (3.5) is reduced to

$$m = \frac{n-1}{n^2} \frac{1-\phi}{\phi} \mathrm{E}[\pi | x + e]. \tag{3.6}$$

It suggests that the total asset under management (AUM) of the fund is

$$M \equiv nm = \frac{n-1}{n} \frac{1-\phi}{\phi} \mathrm{E}[\pi | x + e]. \tag{3.7}$$

Individual and the total fund provisions are proportional to the expected value of trading profits conditional on the fund information disseminated to investors, $\mathrm{E}[\pi | x + e]$. This is a natural consequence of the performance-based fee structure, as it makes the per-unit return of fund delegation proportional to the trading profit. In what follows, we denote the flow-performance sensitivity as $\theta \equiv \frac{n-1}{n} \frac{1-\phi}{\phi}$.

By incorporating the trading strategy in (3.1) and the pricing rule in (2.1), both the trading quantity, x, and the profit margin (i.e., informational advantages), $\mathrm{E}[\delta - p | s]$, are proportional to the realized skill level, $|s|$, making the profit a quadratic function of s. When the fund information is perfectly transparent ($\omega_e = 0$), investors back out the realized skill level from the manager's trading strategy, $s = \frac{x}{\beta}$, so that the conditional expectation leads to the following estimate:

$$\mathrm{E}\left[\pi | x + e\right] = (1 - \lambda\beta)\beta \mathrm{E}[s^2 | s = x/\beta] = \frac{1 - \lambda\beta}{\beta} x^2. \tag{3.8}$$

Together with the total fund provision in (3.7), the above estimate implies that the delegated investment may generate an additional incentive for the manager to control her trading strategy, x. Namely, the manger can positively influence the total fund flow by inflating the trading quantity, as it persuades investors of the manager's high realized skill level.

3.4 Manager's Optimization

By incorporating the total fund flow ([3.7] and [3.8]), the manager's problem in (2.3) is re-written as

$$\max_x \phi \left[sx - \lambda x^2 + \theta \frac{1 - \lambda\beta}{\beta} x^2 \right], \tag{3.9}$$

where the first two terms represents the fee income from the trading profit and are essentially identical to the original Kyle (1985) model. The third term arises due to the fund inflow and is determined by the quadratic size of risky investments. By taking the FOC, the optimal trading strategy involves

$$x = \frac{1}{2} \frac{\beta}{\lambda\beta - \theta(1 - \lambda\beta)} s. \tag{3.10}$$

This strategy is consistent with our guess in (3.1) when the following condition holds:

$$\beta = \frac{1}{2} \frac{\beta}{\lambda\beta - \theta(1 - \lambda\beta)}. \tag{3.11}$$

The solution is summarized by the following lemma:

Lemma 3.1. *When the fund information is transparent ($\omega_e = 0$), the equilibrium trading strategy and the pricing rules are specified by*

$$\beta = \sqrt{(1 + 2\theta)\frac{\omega_u}{\omega_s}}, \ \lambda = \frac{1}{2(1 + \theta)} \sqrt{(1 + 2\theta)\frac{\omega_s}{\omega_u}}. \tag{3.12}$$

As is evident from problem (3.9), the flow-performance sensitivity, θ, captures the manager's additional incentive to control her risky investment arising from the fund delegation, and the equilibrium reduces to the original Kyle (1985) model as $\theta \to 0$. When $\theta = 0$, the manager faces a tradeoff in deciding on her trading intensity: while she seeks to exploit her informational advantage by trading more intensively, it reveals information to the market maker and raises the price impact, cutting into the trading profit. Without fund delegation ($\theta = 0$), the equilibrium β is determined to balance these competing effects. Conversely, $\theta > 0$ brings about the additional incentive: as shown in (3.7) and (3.8), inflating the trading quantity benefits the manager by persuading investors of high $|s|$ and inducing a larger fund flow. When θ increases, the share of the total fund flow becomes larger in the manager's utility, and this last channel becomes more significant. Therefore, β becomes an increasing function of θ. Conversely, the price impact, λ, decreases with θ, as the aggressive trading reveals too much private information to the market maker. From the market maker's perspective, the information is "exaggerated" in that the order size is too large relative to the true $|s|$. Thus, she attempts to partially "undo" the manager's information revelation by incorporating less of it into the price.[10] This mechanism results in a reduced price impact.

This benchmark model and Lemma 3.1 isolate the fundamental mechanism through which the delegated investment structure inflates the manager's investment behavior. The tendency of fund managers to "show off" their skill to attract capital is well discussed in the literature (e.g., Metrick and Yasuda, 2010; Chung, Sensoy, Stern, and Weisbach, 2012; Barber and Yasuda, 2017), raising regulational concerns that fund managers have incentives to exaggerate the performance or quality of their funds.

Although the benchmark model of a transparent fund clearly generates the show-off incentives described above, it is important to ask whether this is a robust implication. In particular, as emphasized by Kyle (1985), a skilled manager may also have incentives to conceal her skill, suggesting a fundamental tension between showing off and hiding. The more general model with opacity, developed in the next section, shows that, contrary to concerns raised by the literature and by policy makers, the manager may in fact prefer to hide her skill from *both* investors and the market.

[10] This is because λ represents the price's reaction to the order flow, $x + u$, rather than its reaction to the private information, s. The latter reaction is measured by $\beta\lambda$ and is monotonically increasing in β and θ.

4 Opaque Funds

This section introduces the fund opacity, $\omega_s > 0$, and demonstrates that belief-driven mltiple equilibria show up regarding the trading strategy of the fund. We discuss economic implications drawn from the equilibrium analyses in Subsection 4.4.

4.1 Delegation under Fund Opacity

Assuming that $\omega_s > 0$, the analysis of an opaque fund largely follows the benchmark case in Section 2. In particular, investors' optimization and the structure of the resulting fund flow are identical to those in (3.6) and (3.7) in the benchmark model. However, due to the noisy revelation of the fund information, investors' conditional expectation about the fund return must be modified as follows:

Lemma 4.1. *Upon observing the fund information, $x + e$, investors' updated expected trading profit is given by*

$$\mathrm{E}\left[\pi|x+e\right] = (1-\lambda\beta)\beta\left(\hat{s}^2 + \hat{\omega}_s\right), \tag{4.1}$$

where the posterior expectation of the (signed) realized skill is

$$\hat{s} = \mathrm{E}[s|x+e] = \xi(x+e), \tag{4.2}$$

with

$$\xi = \frac{\beta\omega_s}{\beta^2\omega_s + \omega_e}, \tag{4.3}$$

and the posterior of the skill variance is

$$\hat{\omega}_s = \mathrm{Var}[s|x+e] = \frac{\omega_s\omega_e}{\beta^2\omega_s + \omega_e}. \tag{4.4}$$

Investors update their belief about the realized skill using $x + e$, where ξ measures how much they rely on the fund information provided by the manager. Holding β constant, ξ is large when investors' prior is unreliable (ω_s is large) and the fund information is transparent (ω_e is small). In what follows, the updated expectation about the manager's realized skill, $\hat{s}$, is referred to as the *reputation*.

4.2 Manager's Optimization

Following the same procedures as the transparent benchmark, the manager's optimization problem in (3.9) is re-written as follows:

$$\max_x \; sx - \lambda x^2 + \theta(1-\lambda\beta)\beta \left[\xi^2(x^2+\omega_e) + \hat{\omega}_s\right]. \tag{4.5}$$

As in the baseline model, the first two terms of equation (4.5) represent the trader's expected trading profit. The third term corresponds to her profits from the fund flow, M, which also increases with $|x|$ because she can potentially inflate her reputation $|\hat{s}|$ by placing a large order (either buy or sell).

The unique solution to (4.5) (given β) is

$$x = F(\beta)s, \tag{4.6}$$

where

$$F(\beta) \equiv \frac{1}{2\left(\lambda - \theta(1-\lambda\beta)\beta\xi^2\right)}. \tag{4.7}$$

$F(\cdot)$ is a nonlinear in β because λ and ξ are functions of β.

4.3 Equilibrium under Opacity

The manager optimally chooses (4.6) given that the market maker and investors believe that she follows strategy (3.1). In equilibrium, their beliefs must be correct, that is, (3.1) and (4.6) must be consistent with each other, requiring the following condition:

$$\beta = F(\beta). \tag{4.8}$$

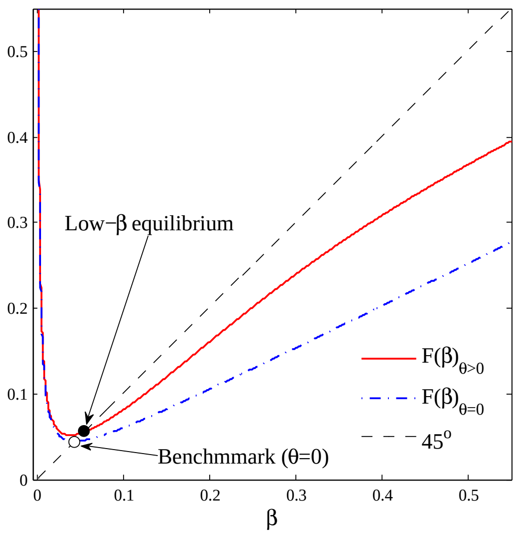
That is, the equilibrium levels of β are determined at the fixed points of $F(\cdot)$. Having identified β, the equilibrium levels of x, p, and m are determined by equations (3.1), (3.2) and (3.6), respectively. In what follows, we first illustrate the determination of equilibrium β through a numerical example, followed by an analytical proposition that formally establishes the equilibria.

Figure 1 illustrates the determination of β for three different levels of fund opacity, ω_e. The intersections of $F(\beta)$ (the solid line) and the 45-degree line yield the equilibrium levels of β. We also plot $F(\beta)$ with $\theta = 0$ (the dash-dotted line) and denote its solution as $\beta_{\theta=0}$ to highlight the original model of Kyle (1985) where fund delegation is irrelevant.

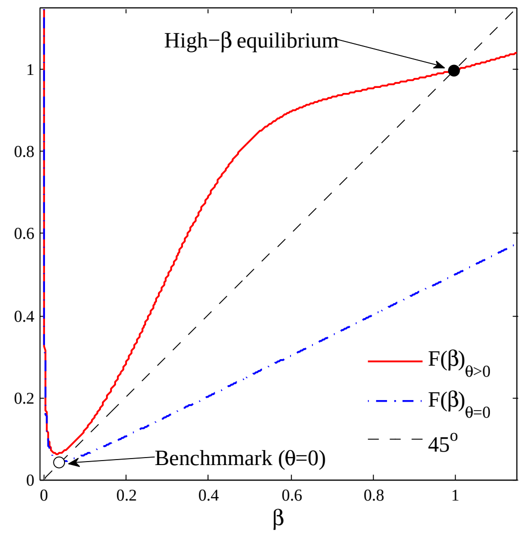
Panel (a) of Figure 1 shows that if ω_e is large, there is a unique "low-β equilibrium" where β is close to $\beta_{\theta=0}$. Panel (b) shows that a large ω_e leads to a unique "high-β equilibrium," in which β is much larger than $\beta_{\theta=0}$. Panel (c) shows that multiple equilibria arise for an intermediate level of ω_e. Specifically, the high-β and the low-β equilibria are stable, as $F(\beta)$ crosses the 45-degree line from above, whereas the one with an intermediate β is unstable, as $F(\beta)$ crosses the line from below.[11] In what follows, we focus only on the stable equilibria.

Impact of fund opacity. To see the intuition behind the different equilibrium patterns presented in Figure 1, note that the fund manager faces two conflicting motives when trading. On the one hand, she seeks to avoid trading too aggressively on her private information, as doing so would move the price excessively and reduce her trading profits. As in Kyle (1985), this motive leads the trader to decrease β. On the other hand, she wishes to trade intensively to signal her high skill to her client investors, aiming to establish a strong reputation and attract a large fund flow. This motive encourages her to increase β. In summary, she faces a tradeoff between "hiding" her skill from the market maker and "showing it off" to her clients. The fund opacity, ω_e, governs

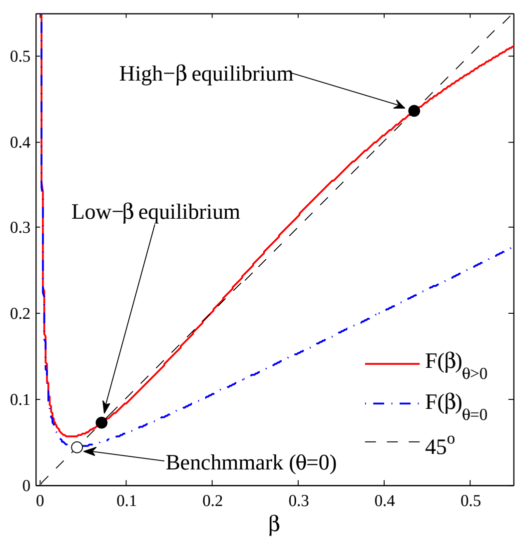
[11]We define an equilibrium to be stable (unstable) if the corresponding β is a stable (unstable) fixed point of $F(\cdot)$, that is, $|F'(\beta)| < 1$ (> 1). A stable equilibrium is robust to a small perturbation to the agents' belief about β, meaning that the economy converges back to the original equilibrium through the best-response dynamics of the agents' beliefs.



(a) Small θ ($\theta = 0.2$) $\Rightarrow$ unique low-β equilibrium.



(b) Large θ ($\theta = 0.55$) $\Rightarrow$ unique high-β equilibrium.



(c) Intermediate θ ($\theta = 0.33$) $\Rightarrow$ multiple high-β & low-β equilibria.

Figure 1: Determination of β

Note: The figure plots $F(\beta)$ for different values of θ. The equilibrium values of β are determined at the fixed points of $F(\cdot)$. The parameter values used in the figures are $\omega_s = 50$, $\omega_\delta = 1$, and $\omega_u = 0.1$.

this tradeoff by influencing the sensitivity of the manager's reputation, $\hat{s}$, to her investment behavior, x. In particular, Lemma 4.1 demonstrates that the fund flow becomes insensitive to x under opacity, as it becomes difficult for investors to estimate the skill based on the fund information, weakening the showing-off motive.

For a opaque fund with a large ω_e (panel (a)), the hiding motive dominates the showing-off motive, leading to a unique low-β equilibrium. In this equilibrium, client investors believe that the manager's trading intensity is relatively low. Knowing this, the manager could inflate her reputation by placing an order larger than what her clients anticipates. However, she would refrain from doing so because it would cause a large price impact, reducing the fund's investment profits. Consequently, the equilibrium β is close to $\beta_{\theta=0}$, meaning that it is a "Kyle-like equilibrium."

For a transparent fund with a small ω_e (panel (b)), the showing-off motive prevails, resulting in a unique high-β equilibrium. In this equilibrium, the manager is willing to move the price substantially, undermining investment profits to appear skilled. Importantly, client investors' learning is not manipulated, as their belief about the manager's trading strategy is correct. Nonetheless, the manager places a large order because doing otherwise would induce her clients to incorrectly estimate that she is less skilled, thereby deteriorating her reputation. She trades aggressively solely to conform to her clients' belief. To emphasize this mechanism, we call this equilibrium the "reputation-driven equilibrium."

For intermediate levels of ω_e (panel (c)), there is no clear dominance between the two motives, hence both the low-β and the high-β equilibria are supported under intermediate fund opacity. Each equilibrium is associated with a self-fulfilling belief of the market maker and investors.

The following proposition and Figure 2 summarize the above discussion and provide an analytical characterization of the equilibria.

Proposition 4.1 (Equilibrium characterization). *There exists a critical value of the fund opacity, $\tilde{\omega}_e = 8\omega_u$, and two thresholds of the flow-performance sensitivity, θ_L and θ_H ($\geq \theta_L$), such that,*

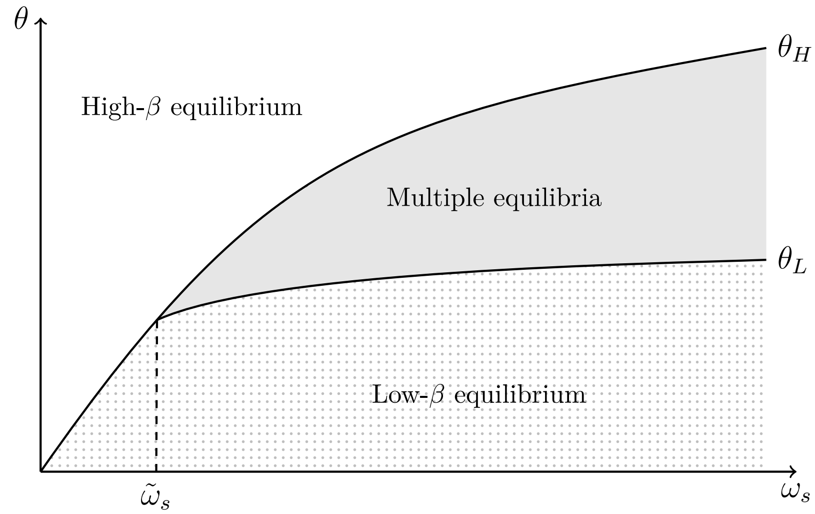


Figure 2: Equilibrium Characterization with Exogenous θ

Note: This figure illustrates the equilibrium states using the reputation concern, θ, and the skill potential, ω_s.

(i) if $\omega_e \leq \tilde{\omega}_e$, then $\theta_L = \theta_H = \theta_0$, and the low-$\beta$ equilibrium uniquely arises when $\theta < \theta_0$, while the high-β equilibrium uniquely arises when $\theta > \theta_0$; and

(ii) if $\omega_e > \tilde{\omega}_e$, then $\theta_H > \theta_L$. $\theta > \theta_H$ leads to a unique high-β equilibrium, $\theta \leq \theta_L$ leads to a unique low-β equilibrium, and $\theta \in (\theta_L, \theta_H)$ leads to multiple equilibria.

Impact of flow-performance sensitivity. Proposition 4.1 and Figure 2 indicate that, for a given fund opacity, ω_e, a weak flow-performance sensitivity supports only the low-β equilibrium, while increases in θ can generate the high-β equilibrium, potentially leading to multiple equilibria.[12] To understand intuition, the tradeoff between hiding and showing-off motives, as explained above, is key. When θ is small, the fund flow's reaction to changes in the expected investment profit becomes weak. In this case, the manager's showing-off motive is less important. Consequently, the hiding motive prevails, prompting her

[12]Lemma **??** in Online Appendix **??** analytically characterizes the upward-sloping θ_H and θ_L in relation to ω_e, as presented in Figure 2.

to trade less aggressively and leading to the low-β equilibrium. Conversely, when θ is large, the fund flow becomes highly sensitive to the fund information, thereby encouraging the manager to show off her skill by trading intensively. This scenario gives birth to the high-β equilibrium.

Overall, the type of equilibrium depends on the combination of θ and ω_e, as illustrated in Figure 2. The tradeoff between the hiding and the showing-off motive tends to balance and generates multiple equilibria when the fund is relatively opaque, and the fund flow is moderately sensitive to the expected investment performance. Otherwise, one of the conflicting motives dominates the other either because of a sufficiently transparent fund, encouraging the manager to show off her skill, or the fund flow being very sensitive or insensitive to the expected performance, altering the importance of the fund flow relative to the investment profit in the manager's optimization.

Equilibrium comparison. Comparing the two stable equilibria, does the manager perform better in one of them? What are the implications for market liquidity, asset prices, and learning about asset-selection skills?

Corollary 4.1 (High-β vs. low-β equilibrium)**.** *In the high-β equilibrium, relative to the low-β equilibrium,*

(i) the market liquidity is higher, that is, λ is lower;

(ii) the manager's expected trading profit given the realized skill, $\mathrm{E}[(\delta - p)x|s] = \frac{\beta\omega_u s^2}{\beta^2\omega_s+\omega_u}$, *is lower;*

(iii) the price informativeness, $\tau \equiv \frac{1}{\mathrm{Var}[\delta|p]} = \frac{\beta^2\omega_s+\omega_u}{\omega_\delta(\beta^2\omega_s+\omega_u)+\omega_u\omega_s}$, *is higher;*

(iv) the price volatility, $\mathrm{Var}[p] = \frac{\beta^2\omega_s^2}{\beta^2\omega_s+\omega_u}$, *is higher; and*

(v) investors' estimate of s is more precise, that is, the posterior variance of s, $\hat{\omega}_s = \frac{\omega_s\omega_u\omega_\delta}{\omega_s(\beta^2\omega_\delta+\omega_u)+\omega_u\omega_\delta}$, *is lower.*

Statement (i) of Corollary 4.1 holds because price impact λ is decreasing in β in the equilibrium. Since the high-β equilibrium emerges only if θ is relatively large, the statement also implies that a higher flow-performance sensitivity leads to higher market liquidity. The intuition is as follows. Relative to the low-β equilibrium, in the high-β equilibrium the manager trades more aggressively, and thus the order flow reflects significant information about s. To incorporate the appropriate amount of information about s into the price, the market maker seeks to offset the exaggerated information in order flow by lowering λ.

The intuition for statement (ii) is closely related to that for statement (i). In the high-β equilibrium, the manager earns a lower trading profit due to a smaller informational advantage over the market maker. This result may appear inconsistent with statement (i) because, ceteris paribus, a lower price impact is typically associated with a higher trading profit. However, the manager's order size $|x|$ in the high-β equilibrium is so large that the *overall* price impact (i.e., λ multiplied by $|x|$) is larger than that in the low-β equilibrium. Consequently, the manager moves the price excessively and earns a lower trading profit in this equilibrium. Due to her aggressive trading, more information about s—which is informative about δ—is incorporated into the price in the high-β equilibrium, supporting statement (iii).

Statement (iv) follows, again, from the manager's aggressive trading in the high-β equilibrium. Specifically, she places a large buy (sell) order when $s > 0$ ($s < 0$), even if $|s|$ is small. This causes a large upward (downward) pressure on p, leading to greater price variability. This result aligns with the insight of Guerrieri and Kondor (2012) that reputation concerns amplify price volatility, although their mechanism differs from ours.[13]

In the high-β equilibrium, the manager reveals more information about s, thereby improving the precision of investors' estimation (statement (v)). This makes the fund flows more responsive to skill, as investors can assess it more accurately.

4.4 Implications

This section discusses economic implications drawn from the equilibrium analyses under opacity.

[13] Guerrieri and Kondor (2012) argue that bond price volatility increases because fund managers demand a premium to offset the risk of reputation damage in the event of default.

4.4.1 Skills, Trading Styles, and Financial Fragility

The self-fulfilling nature of multiple equilibria suggests that financial markets may become vulnerable to non-fundamental factors. A shift in investor beliefs alone, without any structural changes, could cause the market to transition from the standard Kyle-like equilibrium to the reputation-driven equilibrium, characterized by high price volatility and poor trading performance. We interpret this phenomenon as a form of *fragility* in financial markets. Our model suggests that fragility may occur when the flow-performance sensitivity (θ) is moderately high. This result arises when a fund manager covers a large investor base (n), as the total fund flow is increasing in n.

More surprisingly, our model suggests that a lower fund management fee (ϕ) tends to generate equilibrium multiplicity through heightened flow-performance sensitivity. This goes counter to the recent debate on fund managers' incentive compensation and advisory contracts, where payment structures that strongly reward performance are often criticized for encouraging excessive risk-taking. As our model implies, such performance-based compensation can actually stabilize the market by eliminating delegation-driven multiplicity of equilibrium and guiding the economy toward a unique, low-β equilibrium. This is consistent with the empirical finding by Dass, Massa, and Patgiri (2008), who show that high-powered advisory contracts do not induce investment in bubbly stocks but instead reduce such investments, highlighting the stabilizing effect of incentive compensation on financial markets.

Proposition 4.1 yields novel implications connecting the fund opacity, its trading styles, and financial fragility. First, opaque funds ($\omega_e \geq \tilde{\omega}_e$) tend to adopt low-intensity strategies that minimize market impact, prioritizing profits over fund flows, whereas transparent funds ($\omega_e \leq \tilde{\omega}_e$) engage in high-intensity strategies with flashy risk taking, seeking reputation and large fund flows in the industry. The model predicts that these tendencies are more pronounced for funds with very transparent information, as the equilibrium is unique at such levels of ω_e (Proposition 4.1).

These results are somewhat consistent with common observations in the real market. For example, quantitative hedge funds like Renaissance Technologies and Bridgewater are extremely opaque and highly profit-driven, focusing on consistent performance, with benefits from attracting AUM being regarded as secondary to profitability (Schwager, 2012). On the other hand, some transparent funds—such as certain thematic ETFs like ARK Invest or crypto-focused funds—engage in bold, attention-grabbing strategies to attract investor flows. These funds often emphasize visibility and narrative-driven positioning, which can lead to volatile returns and greater flow sensitivity.

The second implication is drawn from the model's multiple equilibria. Namely, the relation between funds' activities and their transparency may become blurry when they become opaque ($\omega_e \geq \tilde{\omega}_e$). Notably, this indeterminacy may lead to financial fragility. Depending on market beliefs, relatively opaque funds may shift between profit-seeking and delegation-driven strategies, creating non-fundamental fluctuations. Our result suggests that even relatively opaque funds may follow the crowd due to market beliefs, hopping between safe and risky strategies.

A notable observation in line with this result is momentum crashes (Daniel and Moskowitz, 2016), where funds engaging in risky and visible momentum trades often switch to safer strategies, generating sharp reversals and amplifying price declines. More generally, the rapid shifts between risk-taking and solid performance-based behavior have contributed to market fragility and volatility, as documented in various studies on economic crises, such as Brunnermeier (2009) on the credit crunch during the Global Financial Crisis in 2007. Importantly, large fundamental shifts that support such transitions are rarely identified (Gennotte and Leland, 1999). Our model does not require such fundamental shocks, as a mere change in beliefs can trigger a shift.

4.4.2 Fund Manager Compensation

Our model provides insights into how skills are reflected in compensation in the investment management industry. As shown in statement (v) of Corollary 4.1, client investors learn s more precisely in the high-β equilibrium than in the low-β equilibrium, as the manager demonstrates her skill by trading more intensively. Consequently, the manager's expected compensation, $\mathrm{E}[\phi y|s]$, be-

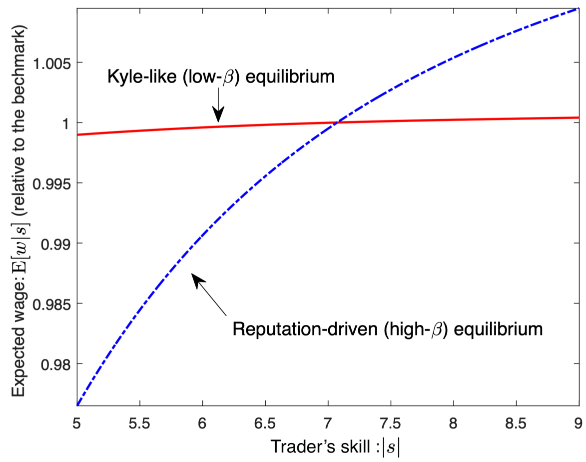


Figure 3: Skill versus Expected Wage

Note: The figure plots the trader's expected wages (relative to the benchmark with $\theta = 0$) at the low-β equilibrium (solid line) and high-β equilibrium (dashed line) for different levels of skill, $|s|$. The parameter values used in the figures are $\omega_s = 50$, $\omega_\delta = 1$, $\omega_{u,0} = 0.1$, $\omega_{u,1} = 60$, and $\theta = \frac{\varphi\beta_1}{2} \approx 0.33$ ($\varphi = 0.6$).

come more sensitive to her realized skill, $|s|$. As an illustration, Figure 3 plots the manager's expected compensation in relation to her skill in the low- and high-β equilibria. In the high-β equilibrium, investors' accurate evaluations result in greater pay inequality: low-skilled managers would receive very low compensation, while high-skilled ones are very highly paid. This result corroborates empirical observations, such as those provided by Philippon and Reshef (2012) and Böhm, Metzger, and Strömberg (2018), highlighting the comparatively large wage inequality in the fund management industry. Célérier and Vallée (2019) attribute its source to the scalability of skill, while inequality in our model arises from managers' incentives to attract fund flows through establishing high reputation.

4.4.3 Skill and Performance Persistence

Do high-skilled managers consistently perform well? Not always: skilled managers may not necessarily outperform less skilled ones due to multiple equilibria. Consider a highly skilled manager and a relatively less skilled one.

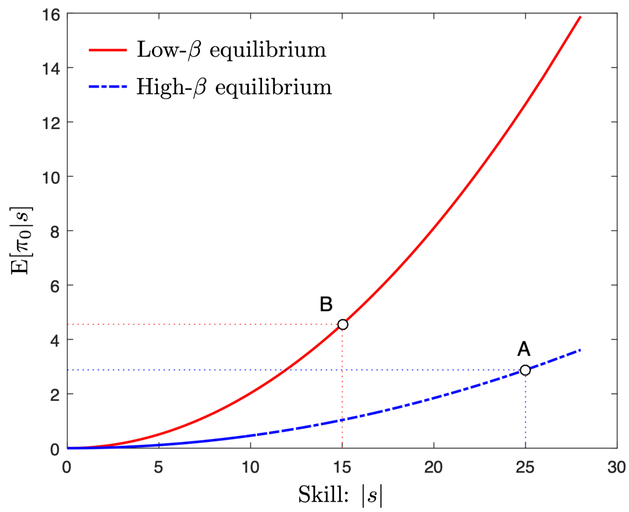


Figure 4: Skill versus Expected Trading Profit

Note: The figure plots the expected trading profit $\mathrm{E}[\pi_0|s]$ at the low-β equilibrium (solid line) and the high-β equilibrium (dashed line) for different levels of skill, $|s|$. The parameter values used in the figure are $\omega_s = 50$, $\omega_\delta = 1$, $\omega_u = 0.1$, and $\theta = \frac{\varphi\beta_1}{2} \approx 0.33$ ($\varphi = 0.6$).

Since trading profits are increasing in managers' skill, $|s|$, the former is expected to achieve higher trading profits than the latter *conditional on* both traders trading at the same intensity, β. However, if their clients believe, for whatever reason, that the skilled manager invest more aggressively than the less skilled one, and managers conform to such beliefs, a high skill level does not translate into performance. This is because the skilled manager, trading at the delegation-driven equilibrium, reveals too much private information to the market maker. Hence, compared to the less skilled manager, who plays the Kyle-like equilibrium, the skilled manager may earn lower expected profits. Figure 4 illustrates such a situation. It plots the expected trading profits (performance) at high- and low-β equilibria with different levels of skill, suggesting that a low-skilled manager in the low-β equilibrium (point B) outperforms a high-skill manager in the high-β equilibrium (point A).

The empirical literature (e.g., Jensen, 1967; Busse, Goyal, and Wahal, 2010) finds that asset managers do not outperform passive benchmarks on average and, in cross-section, those who outperform cannot do so consistently.

While the literature attributes the lack of persistent performance to the lack of skill, our result proposes an alternative explanation. Several studies reconcile the existence of skill with the lack of persistent performance. Berk and Green (2004), for example, argue that skill-based alphas eventually disappear due to competition between investors and decreasing returns to scale at the fund level.[14] Our model contributes to the understanding of this matter by showing that the lack of interplay between manager performance and skill could be the result of self-fulfilling multiple equilibria.

5 Strategic Opacity

This section analyzes endogenous opacity by allowing the manager to control ω_e. We find that she strategically chooses opacity as a commitment device to restrict herself from taking excessive risk. The result

5.1 Setup

There are two periods, $t = 0, 1$. In each period, the trader with skill $|s|$ picks an asset that yields a payoff $\delta_t \sim N(\hat{\delta}_t + s, \omega_\delta)$ at the end of the period, where $\hat{\delta}_t$ is the market maker's prior expectation about δ_t. Each asset lives only for one period. Although the trader picks different assets in $t = 0$ and $t = 1$, both of them are characterized by the same s because s represents her time-invariant skill. In other words, her skill allows her to *systematically* pick an asset such that the market maker's prior expectation about δ_t is biased by $|s|$. We assume that the market is anonymous, meaning that the market maker cannot track which trader selects which assets, while the recruiter can. This assumption, combined with the existence of a continuum of traders with various s, implies that market makers at $t = 1$ cannot update their estimate of each trader's s, whereas the recruiter can.[15] This distinction is critical: it is the market

[14] Pastor, Stambaugh, and Taylor (2015) document increases in skill in finance and attribute the lack of corresponding increases in performance to decreasing returns to scale at the industry level.

[15] This assumption is justified by the anonymity of the market. Each market maker trades with a different trader in each period and cannot track or identify which traders have traded

makers' inability to update the estimate of s that allows the recruiter—who can update it—to potentially make profits by recruiting traders.

Timing. The model unfolds as follows.

1. At the beginning of $t = 0$, each trader draws s independently from $N(0, \omega_s)$, which is her private information. She selects an asset and submits market order x_0 to the market maker for that asset, and liquidity investors' demand $u_0 \sim N(0, \omega_{u,0})$ is realized. The market maker sets a price using order-flow information, $p_0 = \mathrm{E}[\delta_0 | x_0 + u_0]$.
2. At the end of $t = 0$, the asset's payoff δ_0 is realized, and the trader earns the profit, $\pi_0 = (\delta_0 - p_0)x_0$.
3. At the beginning of $t = 1$, each trader meets the recruiter with probability φ. The recruiter observes the past execution price (p_0) and the realized payoff (δ_0) of the asset that the trader has picked in $t = 0$. The trader's x_0 and π_0 are not disclosed. The recruiter updates his estimate of the trader's s, obtaining the posterior mean $\hat{s} = \mathrm{E}[s | p_0, \delta_0]$. The recruiter then employs the trader for a competitive salary:

$$w = \mathrm{E}[\pi_1 | p_0, \delta_0], \tag{5.1}$$

where $\pi_1 = (\delta_1 - p_1)x_1$ is the trading profit in $t = 1$. Note that w is determined by the recruiter's expectation about π_1.[16] This is why the trader cares about the recruiter's perception of her $|s|$, leading to the endogenous reputation benefit. With probability $1 - \varphi$, the trader does not meet the recruiter and remains with her current employer.[17] The

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which assets in the past. Hence, he cannot update traders' skills based on past information. Conversely, recruiters may keep track of traders via direct and individual communications.

[16] An implicit assumption is that the trader encounters a recruiter upon losing her current job. Consequently, she always accepts the salary offered. A trader might find herself in such a situation due to the closure of a trading desk or the discontinuation of a fund, as described by eFinancialCareers ("How Investment Banks are Now Poaching from Floundering Hedge Funds," May 2014). Also, parameter φ can incorporate labor mobility in the finance industry. For example, non-compete clauses and garden leave may reduce φ.

[17] Alternatively, we could introduce the following setting. In the current firm's compen-

trader submits market order x_1 to the market maker. Liquidity investors' demand $u_1 \sim N(0, \omega_{u,1})$ is realized ($\omega_{u,1}$ and $\omega_{u,0}$ can be different). The market maker sets the price $p_1 = \mathrm{E}\left[\delta_1 | x_1 + u_1\right]$.

4. At the end of $t = 1$, δ_1 is realized and π_1 is determined.

The trader's problem at the beginning of $t = 0$ is

$$\max_{x_0} \ \mathrm{E}\left[\pi_0 + \varphi w + (1 - \varphi)\pi_1 | s\right]. \tag{5.2}$$

5.2 Equilibrium

We first solve the problems in $t = 1$ and obtain the salary, w, which is shown to be a quadratic function of the perceived skill, $|\hat{s}|$. Subsequently, we solve the problems in $t = 0$, which are similar to those of Section 2.

5.2.1 Period 1

In $t = 1$, each trader has no reputation concerns. Hence, she simply chooses x_1 to maximize π_1, resulting in the standard Kyle (1985) equilibrium.

Lemma 5.1 (Equilibrium in $t = 1$)**.** *In $t = 1$, each trader's trading strategy is characterized by $x_1 = \beta_1 s$, and the market maker sets the execution price according to $p_1 = \hat{\delta}_1 + \lambda_1 (x_1 + u_1)$, where $\beta_1 \equiv \sqrt{\frac{\omega_{u,1}}{\omega_s}}$ and $\lambda_1 \equiv \frac{1}{2}\sqrt{\frac{\omega_s}{\omega_{u,1}}}$.*

Lemma 5.1 yields the trader's profit, π_1. Then, the *recruiter's* expectation about π_1 determines the trader's salary when she meets the recruiter.

Lemma 5.2 (Trader's salary)**.** *Let $\hat{s}$ and $\hat{\omega}_s$ be the posterior mean and variance of s from the recruiter's point of view. The salary he offers to the trader is*

$$w = \frac{\beta_1}{2}\left(\hat{s}^2 + \hat{\omega}_s\right). \tag{5.3}$$

sation, the proportion α is paid as a fixed wage, and the remaining portion is paid based on performance, π_t, while the recruiter's firm pays the α' portion as the fixed wage. The fixed wage paid by the current firm, $\bar{w}$, is already fixed, so that the trader's expected utility is written as $\mathrm{E}\left[\alpha\bar{w} + (1-\alpha)\pi_0 + (1-\varphi)(\alpha\bar{w} + (1-\alpha)\pi_1) + \varphi(\alpha' w + (1-\alpha')\pi_1)\right]$. Appropriately scaling and rewriting the coefficients of this utility, we obtain a maximization problem virtually identical to equation (5.2), which yields all the results presented in this section.

Lemma 5.2 gives the competitive wage, w, as a quadratic function of $\hat{s}$, providing a microfoundation for the reputation benefit assumed in Section 2.

5.2.2 Period 0

In $t = 0$, the trader knows that she will potentially obtain w in the form of (5.3). Hence, plugging (5.3) into (5.2), her objective function in $t = 0$ is

$$\mathrm{E}\left[\pi_0 + \frac{\varphi\beta_1}{2}\left(\hat{s}^2 + \hat{\omega}_s\right) + (1-\varphi)\pi_1 \middle| s\right]. \tag{5.4}$$

Since π_1 and $\hat{\omega}_s$ are independent of x_0, her optimization problem in $t = 0$ is characterized by (5.5) in the following proposition.

Proposition 5.1 (Endogenous reputation)**.** *In $t = 0$, each trader's maximization problem is identical to* (**??**) *of Section 2, that is,*

$$\max_{x_0} \mathrm{E}\left[\pi_0 + \theta\hat{s}^2 \middle| s\right], \tag{5.5}$$

where parameter θ for reputation concerns is given by

$$\theta \equiv \frac{\varphi}{2}\sqrt{\frac{\omega_{u,1}}{\omega_s}}. \tag{5.6}$$

Proposition 5.1 provides a microfoundation to the quadratic form of the reputation benefit in Section 2. Moreover, θ is endogeneized as a composite parameter (5.6). Not surprisingly, it is increasing in φ. It decreases with the trader's skill potential, ω_s, suggesting that a high-potential trader is less concerned about her reputation. This is due to the *future* trading intensity β_1 and the price impact λ_1: a high-potential trader is likely to possess a high skill, which induces a large price impact in the future as well. Since it is difficult for the trader to exploit her skill in $t = 1$, the competitive wage becomes less responsive to $|s|$, weakening the reputation concern.

As problem (5.5) is effectively the same as (**??**), the equilibrium outcomes resemble Proposition 4.1. However, unlike Proposition 4.1, the impact of the skill potential ω_s on the equilibrium characterization is generalized due to the

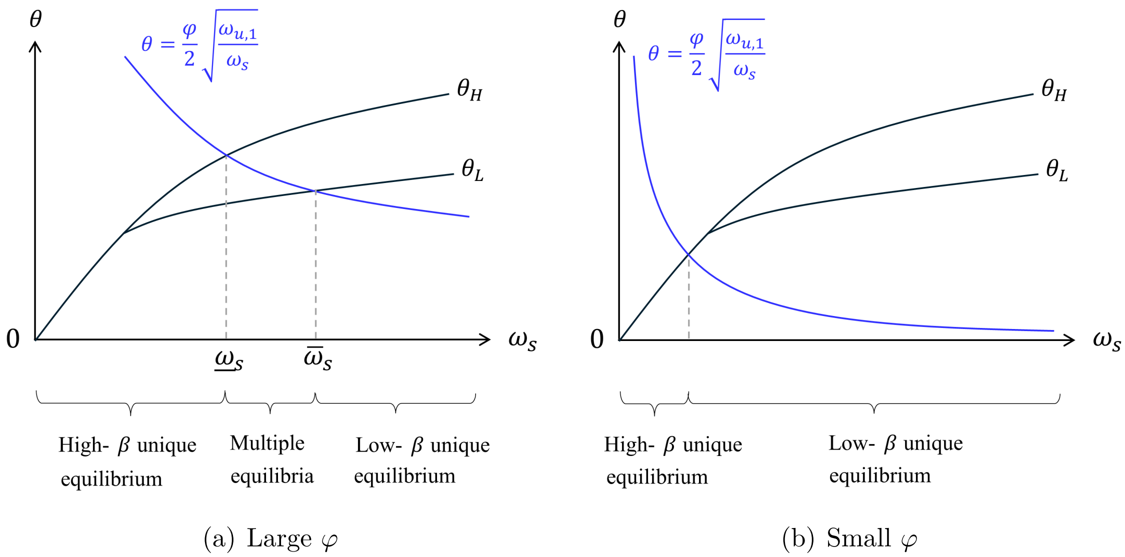

Figure 5: Equilibrium Characterization with Endogenous θ

Note: These figures illustrate the equilibrium states using the reputation concern, θ, and the skill potential, ω_s. The curves represented by θ_H and θ_L are identical to those in Figure 2 and Proposition 4.1, while θ is endogenous and decreasing in ω_s as equation (5.6) suggests.

dependence of θ on ω_s. This is summarized by the following proposition, which is graphically presented in Figure 5.

Proposition 5.2 (Equilibrium in $t = 0$)**.** *The equilibrium at $t = 0$ is characterized as follows.*

1. *If φ is large enough, there are two thresholds for the skill potential, denoted as $\underline{\omega}_s > 0$ and $\bar{\omega}_s$ ($\geq \underline{\omega}_s$), such that*

   (i) for $\omega_s \geq \bar{\omega}_s$, the low-$\beta$ equilibrium is uniquely realized;

   (ii) for $\omega_s \in (\underline{\omega}_s, \bar{\omega}_s)$, both the high- and low-β equilibria arise;

   (iii) for $\omega_s \leq \underline{\omega}_s$, the high-$\beta$ equilibrium is uniquely realized.

2. *If φ is small enough, there is a unique equilibrium. For small ω_s, the high-β equilibrium is realized, while for large ω_s, the low-β equilibrium is realized.*

Since the trader's reputation concern θ is endogeneized as (5.6), which is a decreasing function of ω_s, Figure 5 is obtained by overlaying (5.6) on Figure 2. The first and the second parts of Proposition 5.2 correspond to panels (a) and (b) of Figure 5, respectively. Proposition 5.2 states that multiple equilibria may arise only when labor mobility is sufficiently high (φ is large), i.e., only when the traders have a large reputation concern, θ.[18] Since the case in panel (a) yields richer implications, while panel (b) is qualitatively similar to Kyle (1985), we focus on the case in panel (a), which arises when φ is relatively large.

Panel (a) of Figure 5 shows that, as a result of endogeneizing θ as a function of ω_s, the conditions characterizing the equilibrium types simplify to a single condition solely on ω_s: a large ω_s yields a unique low-β equilibrium, an intermediate ω generates multiple equilibria, and a small ω_s leads to a unique high-β equilibrium.

The intuition is as follows. When the skill potential is high ($\omega_s \geq \bar{\omega}_s$), the trader is more likely to have a large informational advantage over the market maker. This leads to a strong price impact, making it costly to inflate the trading strategy to receive high reputation. Consequently, she is more inclined to adopt profit-seeking strategies, and the Kyle-like equilibrium is uniquely realized. As the skill potential decreases, however, the trader's informational advantage and the price impact tend to diminish, making it easier to prioritize her reputation concerns. Therefore, for intermediate levels of $\omega_s \in (\underline{\omega}_s, \bar{\omega}_s)$, the reputation-driven equilibrium arises and coexists with the Kyle-like equilibrium. If the skill potential declines further ($\omega_s \leq \underline{\omega}_s$), the reputation becomes the dominant factor, and the Kyle-like equilibrium disappears.

5.3 Setup

Before the trading stage in $t = 0$, each trader makes a costly investment to improve her skill potential. This investment could include pursuing an advanced degree in finance, studying for professional exams, or engaging in specialized

[18] Equation (**??**) in Online Appendix **??** provides the definition of the boundary value of φ that classifies cases 1 and 2 in Proposition 5.2.

training to expand her expertise. The trader selects her skill potential, denoted as ω_s^*, by paying the skill-acquisition cost, $C(\omega_s^*) = c\omega_s^*$ with a positive marginal cost $c > 0$.[19] We assume that ω_s^* is not observable to the market makers and the recruiter. Instead, they form a belief about it, denoted by ω_s, and the equilibrium in the trading stage is determined by ω_s following Proposition 5.2. On the equilibrium path, the belief must be consistent with the actual skill potential, $\omega_s^* = \omega_s$. However, at the skill-acquisition stage, the trader cannot influence ω_s, as she does not have a commitment device.[20]

Expected utility of a trader. Given the realized equilibrium in the first trading stage (i.e., β_0), the *ex-ante* expected utility of a trader is characterized by the following lemma:

Lemma 5.3. *For each β_0, the expected utility of a trader at the skill-acquisition stage is given by*

$$V(\omega_s^*, \omega_s, \beta_0) = v(\omega_s, \beta_0)\omega_s^* + \bar{v}, \tag{5.7}$$

where $\bar{v}$ summarizes constant terms unrelated to ω_s^*,[21] and $v(\omega_s, \beta_0)$ is

$$\begin{aligned} v(\omega_s, \beta_0) \equiv &(1 - \lambda_0\beta_0)\beta_0 + (1 - \psi)\frac{1}{2}\sqrt{\frac{\omega_{u,1}}{\omega_s}} \\ &+ \theta\xi^2\left[\gamma^2\beta_0^2 + 2\gamma(1-\gamma)\beta_0 + (1-\gamma)^2\right]. \end{aligned} \tag{5.8}$$

Equation (5.7) indicates that, for each unit of the selected skill potential (ω_s^*), the trader earns the marginal utility of $v(\omega_s, \beta_0)$. The marginal utility in equation (5.8), in turn, consists of three factors. The first two terms represent the maximized trading profit in $t = 0$ and that in $t = 1$ when she does not meet the recruiter. In the first trading round, the equilibrium profit margin of the skill potential is $(1 - \lambda_0\beta_0)\beta_0$, while the profit margin in the second trading

[19]The following analysis does not hinge on the functional form of the cost, as long as it is weakly increasing in ω_s^*.

[20]This setting follows the literature on private information acquisition with an unobservable quality (precision) of the private signal. See, for example, Xiong and Yang (2023).

[21]The formal expression is given by $\bar{v} = \frac{\psi}{4\lambda_1}\left[\xi^2\gamma^2\omega_{u,0} + \xi^2(1-\gamma)^2\omega_\delta + \omega_{u,0}\frac{\gamma\xi}{\beta_0}\right]$

round is identical to that in the standard Kyle (1985) model.[22] The third term is the reaction of the reputation benefit to an increase in the selected skill potential, which emanates from the reputation concerns, θ, and the magnitude of the recruiter's belief updating (see equations [**??**] and [5.3]).

Importantly, $v(\omega_s, \beta_0)$ is characterized by the market's belief about the trader's skill potential (ω_s), as ω_s^* is unobservable to the market. The trader cannot influence ω_s by controlling ω_s^* due to limited commitment. Consequently, the optimality condition for the skill acquisition is characterized by ω_s, as the following discussion elaborates.

Sunspot shock. To formalize the skill-acquisitiom problem based on Lemma 5.3, we introduce an equilibrium-selection device. In particular, following the most common approach, we assume that market participants condition their actions on the realization of an exogenous *sunspot* shock, characterized by the binary random variable $z \in \{0, 1\}$.[23] Namely, when $\omega_s \in (\underline{\omega}_s, \bar{\omega}_s)$, they play the high-$\beta$ equilibrium if a spot appears on the sun ($z = 1$), whereas the low-β equilibrium is realized if no spots are observed ($z = 0$). Accordingly, the high-β equilibrium is realized with probability $\Pr(z = 1) = \rho$, where ρ denotes the exogenous probability of a sunspot. We assume that z is realized at the beginning of the trading stage in $t = 0$, meaning that a trader acquires her skill potential before z is realized. In what follows, the high-β and low-β equilibria are denoted as $\beta_0 = \beta_H$ and β_L, respectively.

By incorporating the equilibrium multiplicity stipulated in Proposition 5.2, the trader's *ex-ante* expected utility from acquiring ω_s^* in equation (5.7) is

[22] In the standard Kyle (1985) model, the optimal trading intensity is $\beta_0 = \frac{1}{2\lambda_0}$, so that the equilibrium profit margin of the information advantage (i.e., the skill in our model) is directly captured by $\frac{\beta_0}{2}$. Our model does not have this feature, as the trading intensity is set by incorporating both the price impact and the reputation concern.

[23] The sunspot equilibrium is proposed by Diamond and Dybvig (1983) as an equilibrium selection device in the context of bunk runs and formalized in the equilibrium analyses by the subsequent studies, such as Cooper and Ross (1998).

rewritten as follows:[24]

$$\mathrm{E}_\beta[v(\omega_s,\beta_0)]\omega_s^* = \begin{cases} v(\omega_s,\beta_L)\omega_s^* & \text{if } \omega_s \geq \bar{\omega}_s, \\ [\rho v(\omega_s,\beta_H) + (1-\rho)v(\omega_s,\beta_L)]\,\omega_s^* & \text{if } \omega_s \in (\underline{\omega}_s, \bar{\omega}_s), \\ v(\omega_s,\beta_H)\omega_s^* & \text{if } \omega_s \leq \underline{\omega}_s. \end{cases} \tag{5.9}$$

E_β denotes the expectation over $\beta_0 = \beta_H$ and β_L in cases of multiple equilibria and incorporates the sunspot shock with $\rho = \Pr(z = 1) = \Pr(\beta_0 = \beta_H)$.

5.4 Equilibrium

Prior to the trading game, each trader solves the following problem by controlling her skill potential:

$$\max_{\omega_s^*} \mathrm{E}_\beta[v(\omega_s,\beta_0)]\omega_s^* - C(\omega_s^*), \tag{5.10}$$

where $\mathrm{E}_\beta[v(\omega_s,\beta_0)]\omega_s^*$ is given by equation (5.9). As the trader cannot control ω_s, she takes the marginal benefit in (5.9) as given.

In the equilibrium, the marginal benefit balances the marginal cost of skill acquisition, as equation (5.11) below illustrates. Otherwise, the trader has an incentive to increase or decrease ω_s^*. Together with the belief-consistency condition, the optimal skill potential ω_s^* is derived as solutions to the following optimality condition with respect to ω_s:

$$\mathrm{E}_\beta[v(\omega_s,\beta_0)] = c. \tag{5.11}$$

Figure 6 plots the marginal benefit (the LHS of [5.11]) in relation to the market's belief about the skill potential, ω_s. Although it is orthogonal to the actual skill potential, ω_s and ω_s^* on the equilibrium path are determined by the intersections of the marginal benefit and the marginal cost curves in Figure 6.

The marginal benefit of skill acquisition globally diminishes as ω_s increases.

[24]We omit the constant term, $\bar{v}$, in the following discussion, as it does not affect the optimal skill potential.

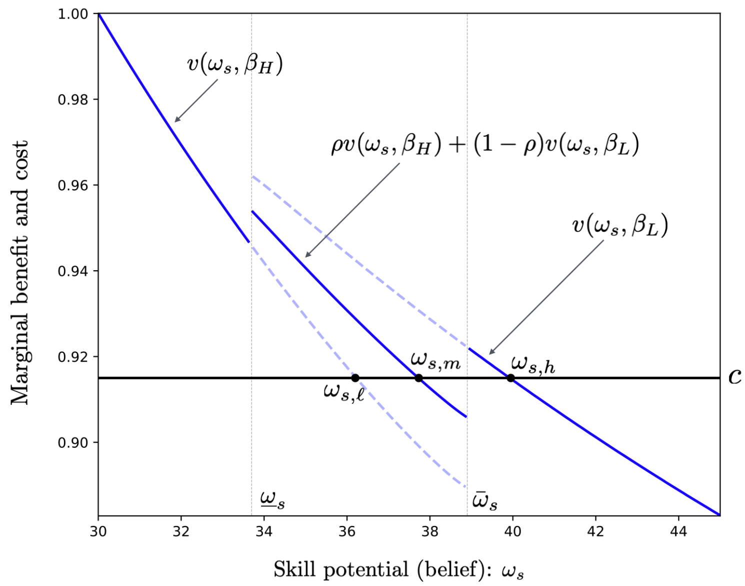

Figure 6: Marginal Benefit and Cost of Skill Acquisition

Note: The figure plots the marginal benefit of skill acquisition, given by $\mathrm{E}_\beta[v(\omega_s, \beta_0)]$, and its marginal cost c against the belief about the skill potential, ω_s. The parameter values are $\omega_\delta = 1$, $\omega_{u,0} = 0.1$, $\omega_{u,1} = 60$, $\theta = \frac{\varphi\beta_1}{2} \approx 0.33$ ($\varphi = 0.6$), and $\rho = 0.5$.

This result is not obvious because there are two channels through which ω_s affects its marginal benefit. In the first channel, when the market maker believes that the trader has acquired a high ω_s, the price impact grows, and the trading intensity diminishes in both trading rounds at $t = 0$ and $t = 1$, thereby reducing the benefit of increasing ω_s^*. Conversely, in the second channel, when ω_s is high, the recruiter faces higher prior uncertainty in estimating the realized skill ($|s|$), and the wage becomes more reliant on the price information (p_0) and the asset payoff in $t = 0$ (δ_0). Hence, the wage becomes more sensitive to the trader's skill, making it more valuable for the trader to acquire a higher skill potential ω_s^*. Although the second channel encourages the trader to increase ω_s^*, the negative effect in the first channel is dominant, forming the downward-sloping marginal benefit curve in Figure 6.

Notably, the marginal benefit curve in the skill-acquisition stage has discontinuities due to the multiple equilibria in the trading stage. When the (belief of) skill potential is sufficiently high or low ($\omega_s \geq \bar{\omega}_s$ or $\omega_s \leq \underline{\omega}_s$), β_0 in the trading stage is uniquely determined. However, when the skill potential turns to intermediate levels, the trader starts incorporating the possibility of multiple equilibria in the trading stage, generating jumps in the marginal benefit of skill acquisition. By focusing on the intermediate levels of $\omega_s \in (\underline{\omega}_s, \bar{\omega}_s)$, we obtain the following result.

Lemma 5.4. *With ω_s being fixed, the marginal benefit of increasing the skill potential is smaller in the high-β equilibrium than in the low-β equilibrium. Therefore, the following inequalities hold, and the marginal benefit curve has discontinuities at $\omega_s = \underline{\omega}_s$ and $\omega_s = \bar{\omega}_s$:*

$$v(\omega_s, \beta_H) < \mathrm{E}_\beta[v(\omega_s, \beta_0)] < v(\omega_s, \beta_L). \tag{5.12}$$

Intuitively, the trader in the high-β equilibrium excessively reveals information to the market maker by trading aggressively. Although the recruiter estimates the trader's skill more precisely and increases the wage's sensitivity to the skill potential, it is dominated by the negative impact of a reduction in the trading profit margin. Therefore, the marginal benefit of increasing the skill potential becomes smaller in the high-β equilibrium than in the low-β equilibrium, leading to inequalities (5.12). In what follows, we elaborate on the discontinuities and derive their implications for skill acquisition.

5.4.1 Middling Skill Acquisition

We first propose "middling" skill acquisition due to the multiple trading-stage equilibria by focusing on $\omega_s \in (\underline{\omega}_s, \bar{\omega}_s)$. We refer to it as middling because the acquired skill level is either higher or lower than what she would acquire if she knew which type of equilibrium would be realized. On the equilibrium path, both the selected skill potential (ω_s^*) and the market's belief about it (ω_s) fall in this region if the marginal skill-acquisition cost is intermediate. In this region, the trading stage involves multiple equilibria (i.e., $\beta_0 = \beta_H$ or β_L), whereas the trader at the skill-acquisition stage cannot influence the equilibrium selection.

Consequently, she determines her skill potential by anticipating the sunspot shock, which leads to the skill acquisition as follows.

Proposition 5.3 (Middling skill acquisition)**.** *The equilibrium skill potential in the presence of multiple trading-stage equilibria is*

(i) lower than the skill potential that she would acquire in the low-β equilibrium; and

(ii) higher than the skill potential that she would acquire in the high-β equilibrium.

Figure 6 illustrates a numerical example; $\omega_{s,m}$ is the ex-ante optimal skill potential, while the ex-post equilibrium is either $\omega_{s,\ell}$ or $\omega_{s,h}$, depending on the equilibrium realization in the trading stage.

The trader invests in skill potential incorporating the possibility of both high- and low-β equilibria. At the trading stage, however, the acquired skill potential turns out to be insufficient if the low-β equilibrium is realized, as its optimal play involves the profit-driven strategy and requires a relatively high asset-selection skill (i.e., $\omega_{s,h}$ is the *ex-post* equilibrium). Conversely, the acquired skill potential is too high if the high-β equilibrium is realized, as the trader takes the reputation-driven strategy that does not require a high skill potential (i.e., $\omega_{s,\ell}$ is the *ex-post* equilibrium).

This result indicates that individuals or firms may invest in skills that seem oddly intermediate or cautious—neither fully specialized nor overly basic. For example, it explains why some traders and finance professionals choose to pursue broader business qualifications instead of more specialized academic paths, allowing them to adapt to a wide range of market situations. Traders also learn intermediate-level coding (e.g., Python) for algorithmic trading or data analysis but stop short of deep computer science expertise. A similar trend can be observed with respect to CFA Levels and quantitative skills among hedge fund managers. These skill acquisitions would be difficult to rationalize in an economy with a single equilibrium but become logical under the equilibrium multiplicity and uncertainty regarding equilibrium realization.

5.4.2 Fragile Skill Acquisition and Skill Diversity

By expanding our scope to the entire range of skill potential (ω_s), the discontinuities in Lemma 5.4 imply that the skill-acquisition problem in (5.10) may have multiple solutions.[25]

For example, when the marginal cost of skill acquisition is moderately small, as Figure 6 illustrates, both the high skill potential ($\omega_{s,h}$) and the intermediate skill potential ($\omega_{s,m}$) become optimal for the trader. Corresponding to $\omega_{s,h}$, the equilibrium in the trading stage is unique with $\beta_0 = \beta_L$, where the trader engages in solid profit-driven trading. Conversely, the trading stage with skill potential $\omega_{s,m}$ involves multiple equilibria, and the selected skill potential is middling, as Proposition 5.3 demonstrates. The symmetric argument holds when the marginal cost is relatively high: the intermediate skill potential and the low skill potential show up as multiple optima. This result can explain at least three realistic phenomena without relying on ad-hoc assumptions on the skill-acquisition environment.

Firstly, the coexistence of traders with different skill levels arises not from heterogeneity in skill-acquisition problems (e.g., different marginal costs), but from the multiple trading-stage equilibria and the possibility of mixed strategies in the skill acquisition stage. This reflects a realistic scenario where, due to advancements in the internet and social media (e.g., online webinars via Zoom and academic courses uploaded on YouTube), opportunities for skill acquisition have become widely and uniformly accessible to all traders, reducing barriers to learning. Despite the increased homogeneity in skill-acquisition costs, we continue to observe heterogeneity in skill levels in the finance industry. Our model offers an explanation for this observation, showing how diverse skill levels can emerge among market participants, without the need to assume inherent differences in their skill acquisition costs.

Secondly, it also captures the possibility of widespread surges in skill acquisition without assuming special conditions for the skill-acquisition environ-

[25] Although the equilibrium multiplicity in the trading stage (i.e., β_0) is determined by the level of ω_s, as Proposition 5.2 stipulates, it is orthogonal to the existence of multiple solutions to the optimal skill acquisition in equation (5.11).

ment. For example, we can explain traders and fund managers in competitive environments (e.g., Wall Street) may collectively strive to acquire high-level skills, such as pursuing advanced degrees, driven by the mixed-strategy equilibrium. This surge in skill acquisition could occur even in the absence of significant changes in external factors, and without the need for heterogeneity in skill-acquisition costs, reflecting industry-wide trends in upskilling and degree inflation, as documented by Philippon and Reshef (2012) and Fuller and Raman (2017).

Finally, discontinuities in the marginal benefit curve add another layer of insight into the effect of the skill-acquisition cost on the equilibrium skill potential. Namely, discontinuities introduce the possibility of significant jumps in the optimal skill level triggered by a small change in the marginal skill-acquisition cost. For instance, when marginal technological advancements slightly reduce the marginal cost of learning, the result could be a sizable upward shift in skill acquisition. Conversely, even a small increase in the cost of learning might cause a large decline in the equilibrium skill level. This discontinuity-driven sensitivity can help explain the historical observations stylized by Philippon and Reshef (2012) that certain external changes lead to abrupt shifts in the distribution of skill levels in the finance industry.

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